

QuantumShield AI - Tools and Methods

ML Libraries Used

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- CatBoost
- XGBoost
- SHAP
- Joblib / Pickle

Models Used

- Random Forest
- XGBoost
- CatBoost (Best Model)

Algorithms Used

- Label Encoding
- Train-Test Split
- Feature Engineering
- Feature Importance
- SHAP Explainability
- Classification

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix

Graphs Generated

- Dataset Preview
- Class Distribution
- Correlation Heatmap
- Histograms
- Box Plot
- Feature Importance

- Model Comparison
- SHAP Summary Plot
- SHAP Bar Plot
- ROC Curve
- Confusion Matrix